

MA201A – Lecture 01 Practice & Homework

Stewart 9e §12.5 & §12.6 · Fully Worked Solutions & Active-Recall Worksheet

1 · Fully Worked Exam & Homework Problems

🔑 P1 · Line through Two Points: $P_0(-1, -4, -4)$ and $P_1(2, 0, -3)$

1. Direction Vector:

$$\vec{v} = \overrightarrow{P_0P_1} = \langle 2 - (-1), 0 - (-4), -3 - (-4) \rangle = \langle 3, 4, 1 \rangle$$

2. Parametric Equations (base P_0 , $t \in \mathbb{R}$):

$$x(t) = -1 + 3t, \quad y(t) = -4 + 4t, \quad z(t) = -4 + t$$

3. Symmetric Equations (all direction numbers $a, b, c \neq 0$):

$$\frac{x+1}{3} = \frac{y+4}{4} = \frac{z+4}{1}$$

4. Segment Equation:

$$\vec{r}(t) = (1-t)\vec{r}_0 + t\vec{r}_1 = \langle -1 + 3t, -4 + 4t, -4 + t \rangle, \quad 0 \leq t \leq 1$$

Verification: $t = 0 \Rightarrow P_0(-1, -4, -4) \checkmark$, $t = 1 \Rightarrow P_1(2, 0, -3) \checkmark$.

🔑 P2 · Line with a Zero Direction Component

Find equations of the line through $P_0(3, -2, 5)$ with $\vec{v} = \langle 4, 0, -7 \rangle$.

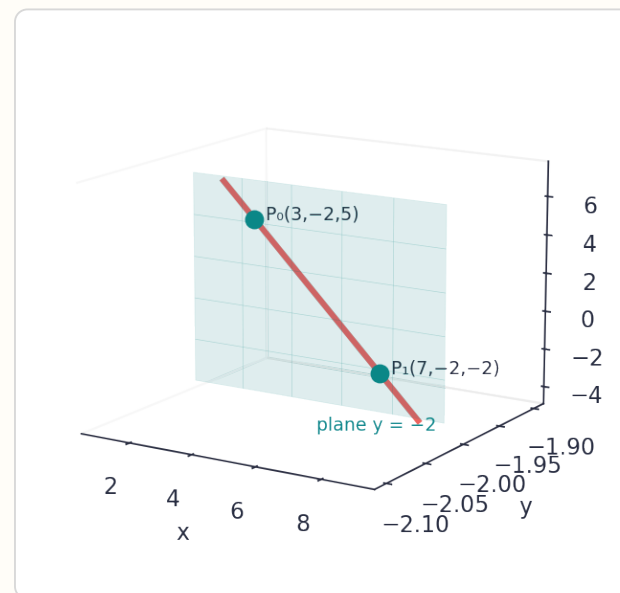
Parametric: $x = 3 + 4t$, $y = -2$, $z = 5 - 7t$ ($t \in \mathbb{R}$).

Since $b = 0$, y is strictly constant across the entire line.

Symmetric Form: Division by zero is forbidden, so state the valid fractions and isolate y :

$$\frac{x-3}{4} = \frac{z-5}{-7}, \quad y = -2$$

Geometry: The entire line is embedded in the plane $y = -2$.



Trapped in plane $y = -2$.

P3 · Classifying a Shifted Quadric by Completing the Square

Classify the surface

$$x^2 - y^2 + 4z^2 - 2x - 4y + 8z + 1 = 0$$

Step 1: Group and complete squares per variable:

$$x^2 - 2x = (x - 1)^2 - 1$$

$$-y^2 - 4y = -(y^2 + 4y) = -[(y + 2)^2 - 4] = -(y + 2)^2 + 4$$

$$4z^2 + 8z = 4(z^2 + 2z) = 4[(z + 1)^2 - 1] = 4(z + 1)^2 - 4$$

Step 2: Substitute and combine constants:

$$(x - 1)^2 - 1 - (y + 2)^2 + 4 + 4(z + 1)^2 - 4 + 1 = 0$$

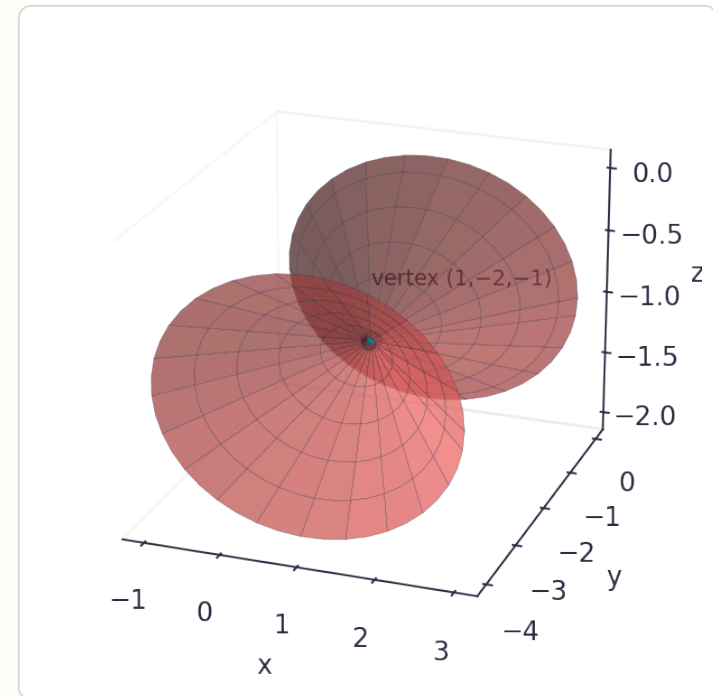
$$(x - 1)^2 - (y + 2)^2 + 4(z + 1)^2 = 0$$

Step 3: Isolate and classify:

$$(y + 2)^2 = (x - 1)^2 + 4(z + 1)^2$$

Right side degree 2, RHS = 0 → **Elliptic Cone**.

Vertex: $(1, -2, -1)$. **Axis:** parallel to y -axis.



Shifted cone vertex $(1, -2, -1)$.

Stewart 9e §12.5 1-5, 13, 17, 23-39

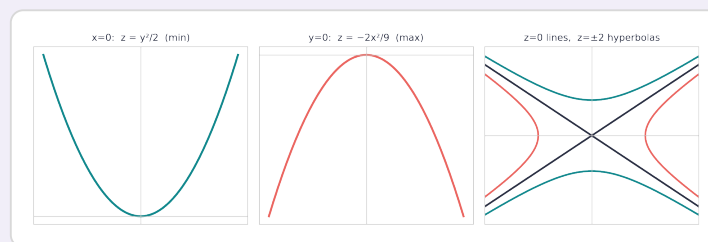
Stewart 9e §12.6 1-7, 23-30, 41, 55

2 · Trace Analysis Master Problem

🔑 P4 · Complete Trace Analysis & Surface Identification — $\frac{z}{2} = \frac{y^2}{4} - \frac{x^2}{9}$

Slice Plane	Trace Equation	Geometric Curve	Physical Behavior at Origin
$x = 0$ (in yz)	$z = \frac{y^2}{2}$	Upward parabola (V at origin)	Along y -axis, origin is a local minimum
$y = 0$ (in xz)	$z = -2\frac{x^2}{9}$	Downward parabola (V at origin)	Along x -axis, origin is a local maximum
$z = 2$ (above origin)	$\frac{y^2}{4} - \frac{x^2}{9} = 1$	Hyperbola opening along y	Cross-sections open north/south
$z = -2$ (below origin)	$\frac{x^2}{9} - \frac{y^2}{4} = 1$	Hyperbola opening along x	Cross-sections flip to east/west
$z = 0$ (in xy plane)	$y = \pm(\frac{2}{3})x$	Two intersecting lines through origin	Asymptotic boundary between level curves

Classification: Because upward curvature along y meets downward curvature along x , the origin $(0,0,0)$ is a **saddle point**. The surface is a **hyperbolic paraboloid** opening along z .



Trace planes flipping axes across $z = 0$.

3 · Test Yourself (Active Recall Worksheet)

Designed for iPad Noteful handwriting with Apple Pencil. Complete before checking the Answer Key.

Q1

A line passes through $A(2, -1, 6)$ and $B(2, 5, -3)$.

- (a)** Find the direction vector $\vec{v}$ and write the parametric equations. **(b)** Write the symmetric equations (identify any zero direction components).
(c) Write the vector equation of the line segment from A to B and verify both endpoints.

Q2

Classify each surface and state its axis of symmetry or opening direction:

(a) $9x^2 - y^2 + 4z^2 = 1$

(c) $4x^2 + y^2 - z^2 = 0$

(b) $x = 2y^2 + 3z^2$

(d) $z = x^2 - 4y^2$

Q3

Consider the equation $x^2 + 4y^2 - 6x + 8y - z + 9 = 0$.

(a) Complete the square and express the equation in standard form.

(b) Identify the surface type and state the coordinates of its vertex.

(c) Write the Maple command to plot this surface centered around its vertex.

Solutions & Answer Key

Check answers only after attempting. Use for self-grading in Noteful.

A1

(a) $\vec{v} = \overrightarrow{AB} = \langle 2 - 2, 5 - (-1), -3 - 6 \rangle = \langle 0, 6, -9 \rangle$.

Parametric: $x = 2, \quad y = -1 + 6t, \quad z = 6 - 9t, \quad t \in \mathbb{R}$.

(b) $a = 0$, so division by zero is illegal:

$$\frac{y+1}{6} = \frac{z-6}{-9}, \quad x = 2$$

The line lives in the vertical plane $x = 2$.

(c) $\vec{r}(t) = \langle 2, -1 + 6t, 6 - 9t \rangle, \quad 0 \leq t \leq 1$.

Check: $t = 0 \Rightarrow (2, -1, 6) = A \checkmark; t = 1 \Rightarrow (2, 5, -3) = B \checkmark$.

A2

(a) One minus sign, $\text{RHS} = 1 \rightarrow$ **Hyperboloid of one sheet**, axis = y .

(b) x is linear, two quad terms same sign $\rightarrow$ **Elliptic Paraboloid**, opens along $+x$.

(c) All degree 2, $\text{RHS} = 0 \rightarrow$ **Elliptic Cone**, axis = z .

(d) z linear, opposite quadratic signs $\rightarrow$ **Hyperbolic Paraboloid** (saddle).

A3

(a) Group terms and complete squares:

$$x^2 - 6x = (x - 3)^2 - 9$$

$$4y^2 + 8y = 4(y^2 + 2y) = 4[(y + 1)^2 - 1] = 4(y + 1)^2 - 4$$

$$(x - 3)^2 - 9 + 4(y + 1)^2 - 4 - z + 9 = 0$$

$$z + 4 = (x - 3)^2 + 4(y + 1)^2$$

(b) **Elliptic Paraboloid**, opening along $+z$, vertex $(3, -1, -4)$.

(c) Maple command:

```
with(plots):
plot3d((x-3)^2 + 4*(y+1)^2 - 4,
       x=0..6, y=-4..2,
       style=patchcontour, axes=boxed);
```